

BIS Compliance Assistant

- Problem Statement ID – **SIH26107**
- Problem Statement Title – **AI-Powered Intelligent Assistance for Indian Standards and BIS Services for Industries and Consumers**
- Theme – **Smart Automation**
- PS Category- **Software**

Idea Title — Proposed Solution

Proposed Solution

- Understands BIS queries typed in plain, everyday language no need to know IS codes.
- Maps the user's product or requirement directly to the correct Indian Standard.
- Lays out the certification, testing, documentation and compliance steps that apply.
- Converts complex BIS standards into simple, easy-to-understand explanations.
- Backs every answer with cited BIS documents/sections to avoid hallucination.

How It Addresses the Problem

Complex info → simplified

AI explains standards in plain language.

Hard to identify standard →

Automated Product → Standard mapping.

Scattered sources → unified

One conversational interface for all BIS info.

Testing/lab confusion →

Standard-wise testing & lab discovery.

Manual checking → AI checklist

Source-backed, RAG-grounded responses.

Innovation & Uniqueness

Product-to-Compliance Mapping

Product → Standard → Cert → Test → Lab → Action.

RAG + Knowledge Graph

Contextual, evidence-based answers.

AI Compliance Gap Analysis

Flags missing documents, tests or requirements.

Standards Change Intelligence

Tracks amendments & notifications over time.

Personalized Roadmap

"What applies?" → "Why?" → "What's missing?" → "Next step?"

Technologies Used

Frontend

React.js, Tailwind CSS, HTML5/CSS3, TypeScript

Backend

Python, FastAPI, REST APIs

AI / ML

LLM, RAG, NLP, Embeddings, Semantic Search, Reranking

Knowledge & Data

PostgreSQL, pgvector/Qdrant, Neo4j Knowledge Graph

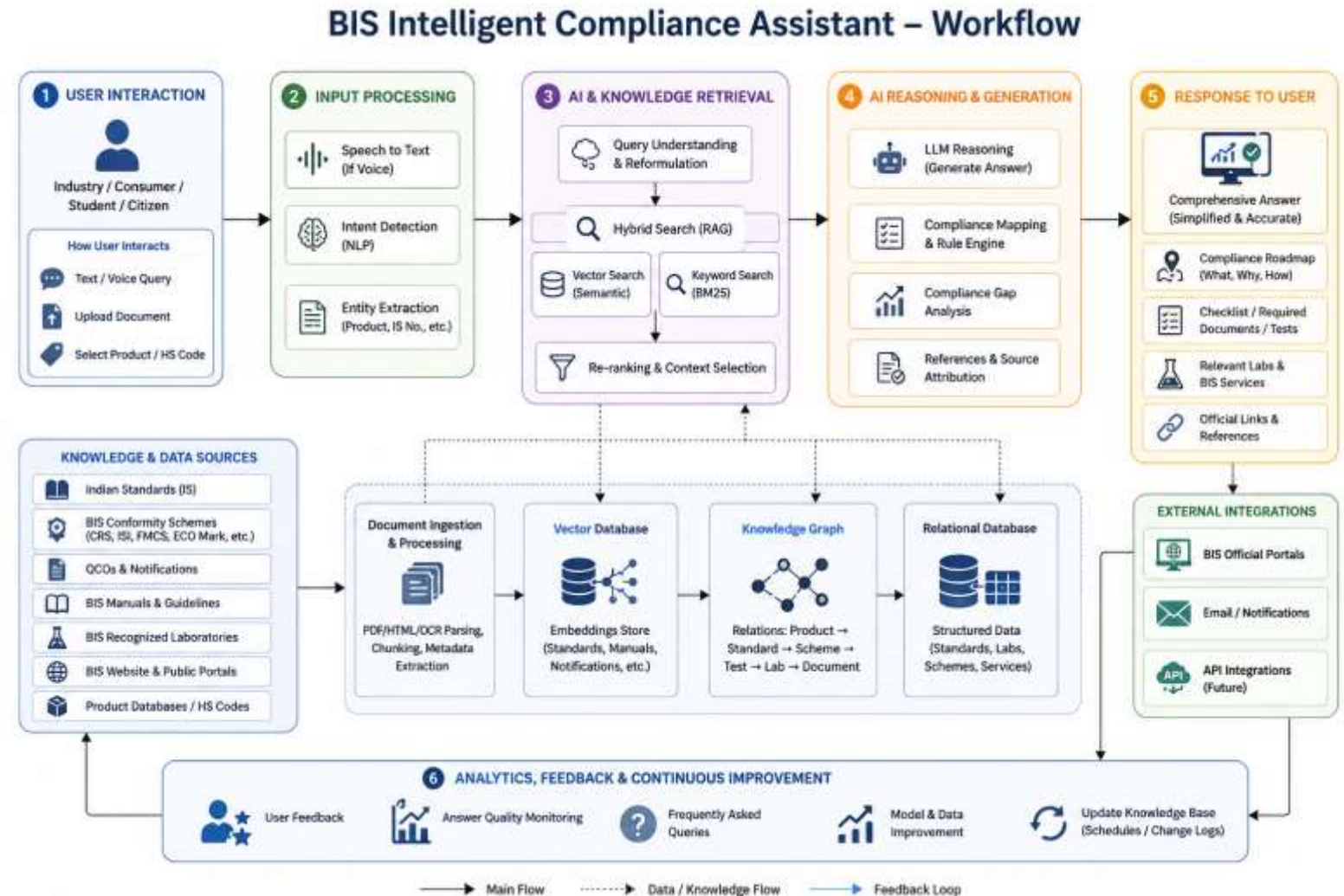
Document Processing

PyMuPDF, OCR, Chunking & Metadata Extraction

Deployment & Security

Docker, AWS/Azure/GCP, JWT & RBAC, Encryption

Workflow / Architecture Diagram



Feasibility of the Idea

Technical Feasibility

Ready-made LLM APIs, vector DBs & embedding models.

MVP Feasibility

Start with 3–5 product categories, not full BIS ecosystem.

Scalability

Prototype → Selected Standards → Wider BIS Services → National.

Web Architecture

Web-based build enables rapid development & deployment.

Potential Challenges

Data Availability — Complete, continuously updated BIS data may lack a single public API.

AI Hallucination — Incorrect AI-generated compliance info could mislead users.

Changing Standards — Standards, amendments & notifications change over time.

Data Access & Licensing — Not every document is freely downloadable/redistributable.

Data Privacy — Uploaded product documents may hold confidential info.

Strategies to Overcome

Focused, Curated Dataset — Begin with verified BIS information; expand gradually.

Retrieval-Augmented Generation — Responses grounded in retrieved BIS sources, not memory.

Version Control — Track standard number, version, amendment & update date.

Source Verification — Show Answer → BIS Source → Document → Section/Page.

Data Security — Encryption, access control & secure document processing.

Target Audience

Industries & MSMEs

Consumers

Testing Labs

BIS / Government

Students & Researchers

Social Benefits

- Makes BIS information accessible to diverse users
- Enables multilingual, simplified interaction
- Improves consumer awareness of standards & quality
- Supports students & researchers

Economic Benefits

- Reduces time spent searching/interpreting requirements
- Cuts avoidable compliance-related costs
- Supports MSMEs in certification preparation
- Accelerates product compliance & market readiness

Environmental Benefits

- Encourages adherence to quality & efficiency standards
- Can reduce defective-product waste
- Supports longer-lasting, better-quality products

National Benefits

- Improves accessibility & usability of India's standards ecosystem
- Strengthens adoption of conformity-assessment processes
- Reduces repetitive queries to BIS / government

Reference & Research Sources

Bureau of Indian Standards — Official Portal

bis.gov.in — standards catalogue, certification & licensing information.

IS Standards / e-BIS Search Portal

Product-to-Indian-Standard lookup and certification schemes (ISI, CRS, Hallmarking).

RAG & Knowledge-Graph Research

Open literature on Retrieval-Augmented Generation and graph-based knowledge representation for grounded AI answers.

MSME & Industry Compliance Guidelines

Ministry of MSME resources on certification and testing requirements for industries.

Add specific reference links / citations (papers, datasets, portals) used for your prototype here.